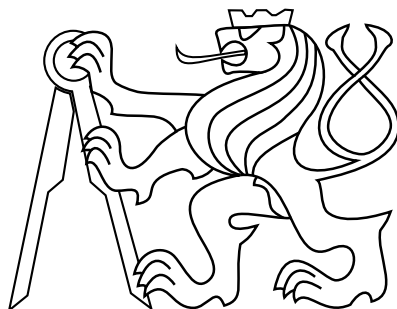


CZECH TECHNICAL UNIVERSITY IN PRAGUE

FACULTY OF ELECTRICAL ENGINEERING
DEPARTMENT OF CYBERNETICS



Classification of journalistic quality of articles

Bachelor's Thesis

Olga Pimenova

Prague, May 2025

Study programme: Open Informatics
Branch of study: Artificial Intelligence and Computer Science

Supervisor: Ing. Jan Šedivý, CSc.

Acknowledgments

I sincerely thank my academic supervisor, Ing. Jan Šedivý, CSc., for his expert guidance, patience, and encouragement. His profound knowledge of advanced technologies and dedication to academic excellence significantly shaped the direction and quality of my research.

I am equally grateful to my workplace supervisor, Radek Tomšů, for his practical insights and mentorship during the implementation phase. His feedback effectively bridged theoretical concepts with real-world applications, enhancing the project's outcomes.

Special thanks to the Seznam.cz recommendation systems team for providing technical resources, collaborative opportunities, and industry perspectives that strengthened the experimental aspects of this thesis.

Finally, I owe immense gratitude to my family for their unwavering emotional support, sacrifices, and belief in me. Their encouragement during challenging moments was crucial to completing this journey.

Swap this for the Thesis Assignment, when you have it from the department.

Declaration

I declare that this thesis was independently developed and that all sources of information used are cited in accordance with the university's ethical guidelines for thesis preparation. Additionally, I disclose that I utilized the AI tool ChatGPT to assist in translating certain sentences from my native language into English and to rephrase them in a style more appropriate for academic writing.

Prague, May 2025

Abstract

TODO:

Keywords Unmanned Aerial Vehicles, Automatic Control

Abstrakt

TODO:

Klíčová slova Bezpilotní Prostředky, Automatické Řízení

Abbreviations

TP True Positive

FP False Positive

FN False Negative

BERT Bidirectional Encoder Representations from Transformers

Insufficient Articles Class 0

Short but Sufficient Articles Class 1

Standard Articles Class 2

High-Quality Articles Class 3

BROWCzech Base-sized Roberta using Wordpiece tokenization for Czech

RetroMAE Retrieval-Oriented Masked Autoencoder

DeBERTaV3 Decoding-enhanced BERT with Disentangled Attention

ELECTRA Efficiently Learning an Encoder that Classifies Token Replacements
Accurately

RTD Replaced Token Detection

MLM Masked Language Modeling

CE Cross-Entropy Loss

MSE Mean Squared Error

Contents

1	Introduction	1
1.1	Motivation	1
1.2	Problem Formulation	2
1.3	Structure of the Thesis	2
1.4	Related Work	3
1.4.1	Quality Assessment Frameworks in Journalism	3
1.4.2	Computational Approaches to Text Quality Assessment	3
1.4.3	Machine Learning for Classification of Media Content	3
1.4.4	Multi-class Classification in Natural Language Processing	3
1.4.5	Linguistic Features for Quality Assessment	4
1.4.6	Research Gaps	4
1.5	Contributions	4
1.5.1	Development of a Robust Classification System:	4
1.5.2	Innovative Handling of Class Imbalance and Overfitting:	5
1.5.3	Comprehensive Feature Engineering for Journalistic Quality:	5
1.5.4	Comparative Analysis of Pre-trained Language Models:	5
1.5.5	Advanced Model Testing and Superior Performance:	5
1.5.6	Innovations in Data Collection and Preprocessing:	5
1.5.7	Establishment of a Human Editor Benchmark:	5
2	Data Collection and Analysis	7
2.1	Data Source	7
2.2	Data Collection Timeline and Challenges	7
2.3	Data Cleaning and Filtering	7
2.4	Annotation Process	8
2.5	Dataset Statistics	8
2.5.1	Dataset Size and Splits	8
2.6	Data Analysis	9
2.6.1	Article length	9
2.6.2	Length by label	9
2.6.3	Unique-token probability density	10
3	Evaluation Metrics	11
3.1	Limitations of Accuracy	11
3.2	Precision	11
3.3	Recall	12
3.4	F1 Score	12
3.5	Weighted F1 Score	13
3.5.1	Why use weighted F1?	13

3.6	Confusion Matrix	13
3.7	Business Relevance	14
3.7.1	Cost of Classification Errors	15
4	Methodology	16
4.1	Editor Analysis and Upper Bound	16
4.1.1	Why Human Editors Are Important	16
4.1.2	How We Studied Editor Agreement	16
4.1.3	What We Learned About Editor Agreement	16
4.1.4	Setting the Upper Bound	16
4.2	Baseline Model Performance	18
4.2.1	Random Classifier	18
4.2.2	Most Frequent Class Classifier	18
4.2.3	Discussion	18
4.3	Performance of Pre-trained Language Models	20
4.3.1	Theoretical Foundations	20
4.3.2	Evaluation of Multiple Pre-trained Models	20
4.3.3	Model Selection	24
4.4	Classifier Architecture Design	25
4.5	Loss Function Engineering	26
4.5.1	Initial Approach: Cross-Entropy Loss	26
4.5.2	Shortcomings of Using CE Alone	26
4.5.3	Addressing Class Imbalance: Focal Loss vs. Class Weighting	27
4.5.4	Incorporating an Ordinal Component: CE + MSE	28
4.5.5	Final Results and Advantages of the Hybrid Loss	30
4.6	Training Methodology	30
4.7	Training Stages	31
4.7.1	Hyperparameter Configuration	31
4.7.2	Validation and Early Stopping	31
4.8	Addressing Class Imbalance	31
4.8.1	Impact of Imbalance on Training	31
4.8.2	Class Weighting Strategy	31
4.9	Feature Engineering	31
4.10	CatBoost	31
4.11	Per-Class Performance Analysis	31
4.12	Error Analysis	31
5	Experiments and Results	32
5.1	Experimental Setup	32
5.2	Model Comparison	32
5.3	Feature Importance Analysis	32
6	Discussion, Conclusion and Future Work	33
6.1	Discussion	33
6.2	Conclusion	33
6.3	Future Work	33
7	Conclusion	34

8	References	35
A	Appendix A	36

1 Introduction

1.1 Motivation

The digital content revolution has transformed journalism into a high-velocity industry where platforms must simultaneously manage exponential content growth and maintain rigorous quality standards. Seznam.cz, serving over 5 million daily users, currently relies on human editors to manually assess only 20% of submitted articles - an unsustainable practice given the platform's 50% annual content growth rate. This manual process introduces significant challenges:

- **Scalability limitations:** With article submissions increasing by 50% annually, human editors cannot maintain pace without compromising review depth
- **Consistency issues:** Inter-editor agreement analysis reveals 20% variance in quality ratings for identical content
- **Quality degradation risks:** Unreviewed articles risk platform credibility through potential misinformation or poorly structured content

These limitations highlight the need for more efficient and consistent methods for evaluating journalistic quality. Automated quality classification addresses these challenges by providing:

- Real-time assessment for 100% of submissions
- Consistent application of editorial guidelines
- Significantly reduce the workload on human editors
- 24/7 operational capacity without human fatigue
- Data-driven insights for continuous quality improvement

Therefore, the primary motivation for this research is to develop an automated system capable of classifying the journalistic quality of articles, mimicking the complex decision-making process of experienced human editors.

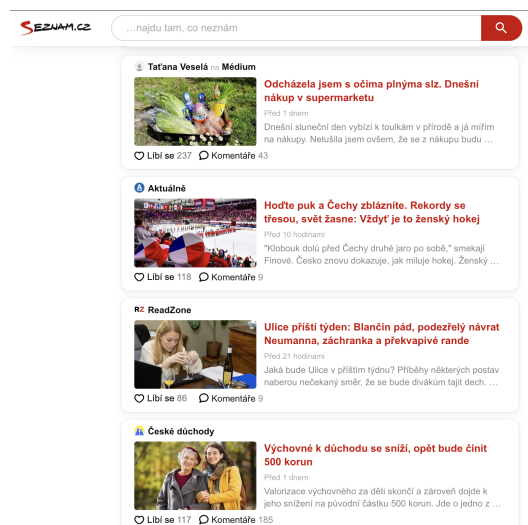


Figure 1.1: Main page Seznam.cz

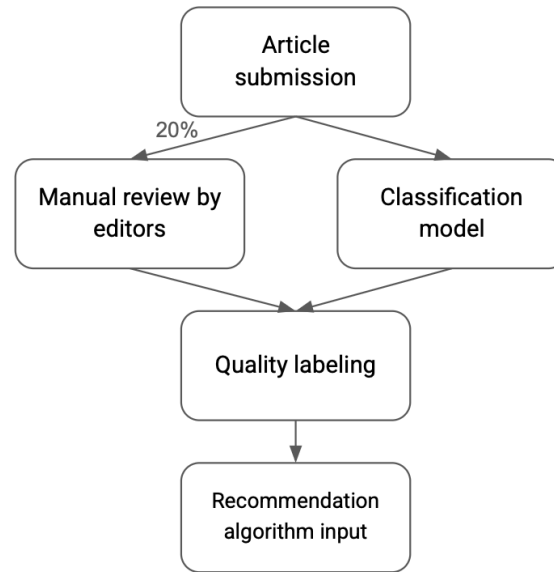


Figure 1.2: Current Editorial Workflow

■ 1.2 Problem Formulation

The core problem addressed in this thesis is the multi-class supervised classification of journalistic articles based on their quality. Our goal is to develop a computational model that automatically assigns an article to one of four predefined quality categories, reflecting the standards used by editors at Seznam.cz.

- Class 0 (Insufficient): Low-quality content that lacks structure and depth.
- Class 1 (Short but Sufficient): Lacks depth and analysis, often recycling known information with limited sources.
- Class 2 (Standart): Reliable news articles with at least five paragraphs and two references.
- Class 3 (High-Quality): Long, detailed, well-structured articles with visuals, subheadings, and references.

In the context of this thesis, **”journalistic quality”** is defined by a combination of factors that human editors consider. These include textual properties like readability, complexity, and style; structural attributes such as organization, paragraphing, and the use of headings; content characteristics like depth of analysis, factual accuracy, and source referencing; and metadata signals like the publishing channel. The aim is to create a model that learns to identify the patterns associated with these factors to predict the appropriate quality category. A significant challenge within this problem is the inherent imbalance in real-world data, where lower-quality categories are often much more frequent than the highest-quality ones.

■ 1.3 Structure of the Thesis

■ 1.4 Related Work

The automated classification of journalistic quality is a multidisciplinary challenge that integrates natural language processing (NLP), machine learning (ML), and media studies. This section critically reviews existing literature across these fields, highlighting significant contributions and identifying gaps that this thesis addresses, particularly in the context of the Czech language media environment and platforms like Seznam.cz.

■ Quality Assessment Frameworks in Journalism

Traditional journalistic quality assessment relies on human evaluation guided by professional standards. **lacy2015** established eight core dimensions of journalistic quality, including accuracy, context, and sourcing diversity. While comprehensive, their framework was designed for manual evaluation, limiting its applicability to automated systems. Recent work by **calvo2024** extends these principles to AI-assisted journalism, emphasizing transparency, accountability, and ethical considerations as critical criteria for maintaining quality in digital news production. Their study highlights that AI can enhance efficiency but risks reducing nuance and context, necessitating robust evaluation frameworks.

■ Computational Approaches to Text Quality Assessment

Automated text quality assessment provides foundational insights for journalistic applications. **pitler2008** demonstrated correlations between syntactic features and human perceptions of writing quality, using metrics like sentence complexity and coherence. However, their approach was general and did not address the specific stylistic and ethical requirements of journalistic texts. More recent advancements, as noted in a 2024 article by **quantum2024**, leverage NLP techniques such as grammar analysis, sentiment analysis, and text cohesion metrics to evaluate content quality automatically. These methods offer promising tools for journalism but require adaptation to domain-specific standards.

■ Machine Learning for Classification of Media Content

Machine learning has significantly advanced media content classification. **potthast2018** developed methods to detect style-based bias in news articles, focusing on political orientation. Their work illustrates the potential of ML to identify subtle textual patterns but does not directly address overall quality. Similarly, **gonzalez2021** applied BERT-based architectures to classify news articles by credibility, achieving accuracy rates above 85%. However, their binary classification approach (credible vs. non-credible) does not capture the nuanced spectrum of journalistic quality, which often requires multi-class models.

■ Multi-class Classification in Natural Language Processing

Multi-class classification in natural language processing (NLP) is a critical technique for categorizing text into multiple predefined classes, making it highly relevant for assessing the quality of journalistic content across various dimensions, such as accuracy, depth, and balance. A comprehensive review by **minaee2020** examines over 150 deep learning models applied to text classification tasks, including sentiment analysis, news categorization, and question answering. The authors highlight the effectiveness of transformer-based architectures, such as BERT and its variants, which leverage large-scale pretraining and fine-tuning to achieve

state-of-the-art performance in complex multi-class classification problems. These models are particularly promising for evaluating journalistic quality, as they can capture semantic nuances and contextual relationships within news articles. The review also summarizes over 40 popular datasets, some of which focus on news categorization, providing valuable benchmarks for developing quality assessment systems.

Similarly, **schmidt2019** introduced a multi-class approach for classifying Wikipedia article quality using document embeddings and content features. Their method, which achieved high accuracy in distinguishing quality levels, demonstrates the potential of semantic representations for quality assessment, adaptable to journalistic contexts.

■ Linguistic Features for Quality Assessment

Identifying linguistic features that correlate with quality is a key research area. **schmidt2019** utilized document embeddings to capture semantic content, combined with features like section length and readability scores, to assess article quality. In journalism, similar features—such as Flesch-Kincaid readability, lexical diversity, and syntactic complexity—can indicate quality but must be tailored to journalistic standards. The Czech language, with its rich morphology and inflectional system, presents unique challenges for NLP tools, as standard models designed for English may underperform.

■ Research Gaps

Despite progress, several gaps persist in the literature. First, most studies focus on English-language content, with limited exploration of languages like Czech, where morphological complexity affects NLP performance. Second, many approaches rely on binary classification, which oversimplifies the multi-dimensional nature of journalistic quality. Third, few studies address the specific needs of news aggregation platforms like Seznam.cz, where high content volume and diversity demand scalable, language-specific solutions. Finally, ethical concerns, such as bias in AI models and the loss of human judgment, remain underexplored in automated quality assessment.

This thesis aims to address these gaps by developing a multi-class classification model tailored for the Czech language media environment. By integrating textual analysis, metadata, and Czech-specific linguistic features, this work builds on existing research while tackling its limitations, particularly for news aggregation platforms.

■ 1.5 Contributions

This thesis makes several significant contributions to the field of automated journalistic quality assessment, particularly in the context of the Czech language and platforms like Seznam.cz. These contributions are outlined below:

■ Development of a Robust Classification System:

- Introduced a DeBERTaV3-based classification system for categorizing journalistic articles into four quality levels: Insufficient, Short but Sufficient, Standard, and High-Quality.
- Achieved a weighted F1 score of 0.77, closely approaching the human editor agreement upper bound of 80%, demonstrating the system's effectiveness in mimicking human editorial judgments.

■ Innovative Handling of Class Imbalance and Overfitting:

- Addressed the challenge of class imbalance (e.g., only 0.9% of articles are High-Quality) through sequential training, a hybrid loss function combining Cross-Entropy and Mean Squared Error, and class weighting ([1.0, 0.5, 1.0, 20.0]).
- These strategies effectively balanced model performance across all classes, particularly improving recall for the minority High-Quality class.

■ Comprehensive Feature Engineering for Journalistic Quality:

- Developed a feature set grounded in journalistic quality dimensions (Diversity, Transparency, Attractiveness, and Independence).
- Incorporated linguistic complexity features (e.g., syntactic complexity, lexical diversity), structural composition features (e.g., section organization, citation quality), readability metrics adapted for Czech, and metadata features (e.g., channel_id, publication timing).
- Utilized Latent Semantic Analysis (LSA) to reduce text embeddings to a single dimension, streamlining quality assessment.
- Introduced new features such as readability, manipulativity, and formality, addressing gaps in previous research.

■ Comparative Analysis of Pre-trained Language Models:

- Evaluated three pre-trained language models (BrowCzech, RetroMAE, DebertaV3 base and DebertaV3 large) for Czech journalistic content.
- Identified DebertaV3 large as the superior model due to its relative position embeddings and bilingual capabilities, providing insights for model selection in similar tasks, especially in low-resource languages.

■ Advanced Model Testing and Superior Performance:

- Tested multiple models, including a neural network with Bert-like models embeddings and a CatBoost classifier.
- Achieved the highest F1 score of 0.77 with the DebertaV3 large + CatBoost model.

■ Innovations in Data Collection and Preprocessing:

- Created refined datasets (e.g., editor-reviewed, unchecked) and cleaned text data by removing emojis and HTML tags.
- Highlighted the importance of article length as a quality indicator, with High-Quality articles averaging 1542.5 words compared to 486.5 for the majority class.

■ Establishment of a Human Editor Benchmark:

- Conducted a detailed editor agreement study, setting a realistic upper bound of 80% for automated systems.
- This benchmark provides a target for model performance and validates the complexity of journalistic quality assessment, ensuring that automated systems are evaluated against a feasible standard.

These contributions collectively advance the state-of-the-art in automated journalistic quality assessment, offering a scalable and effective solution for platforms like Seznam.cz while addressing technical challenges such as class imbalance, overfitting, and language-specific nuances. The thesis also lays a foundation for future research by documenting extensive experiments, baseline comparisons, and error analyses.

■ 2 Data Collection and Analysis

■ 2.1 Data Source

The data for this research originates from the internal systems of Seznam.cz, a major Czech online platform. The dataset comprises articles published on the platform, along with associated metadata and quality assessments provided by the editorial team. For the purpose of this thesis, we primarily utilized the following information for each article:

- **Title:** The headline of the article.
- **Content:** The full text body of the article.
- **Channel ID:** An identifier for the specific source or section within Seznam.cz where the article was published.
- **Entity type:** Type of entity (e.g., article, video, etc.)
- **Published date:** Date when the content was published
- **Content type:** Type of content (e.g., news, blog, etc.)
- **Annotated Quality Correction:** The quality label assigned by a human editor, serving as the ground truth for our classification task.

■ 2.2 Data Collection Timeline and Challenges

Data collection for this study began in July 2023 but faced significant challenges due to data transmission issues between departments. These issues affected the integrity of the dataset during the initial phases of collection. To illustrate the impact of these challenges, a plot (Fig. 2.1) of the number of labeled articles per day reveals inconsistencies in labeling frequency.

These problems were resolved after **August 30, 2024**, when accurate recording of confirmations began.

Additionally, discussions with editors revealed that the evaluation process evolved significantly over time. After **August 30, 2024**, more detailed training was introduced for editors, leading to cleaner and more consistent data. These improvements enhanced the quality of annotations, making the data collected after this date particularly robust.

As a result, the dataset used in this study benefits from a refined evaluation process and improved editor training, ensuring higher reliability and consistency in the annotations used for training and evaluating the classification model.

■ 2.3 Data Cleaning and Filtering

To prepare the dataset, the following steps were taken:

- (i) Articles with fewer than 50 words were excluded to ensure sufficient content for classification.
- (ii) Text was cleaned to remove emojis, HTML tags, and other non-text elements, focusing on title and main content.
- (iii) Only articles were included, excluding other entity types like videos or images.

These measures ensured a clean and relevant dataset for journalistic quality classification.

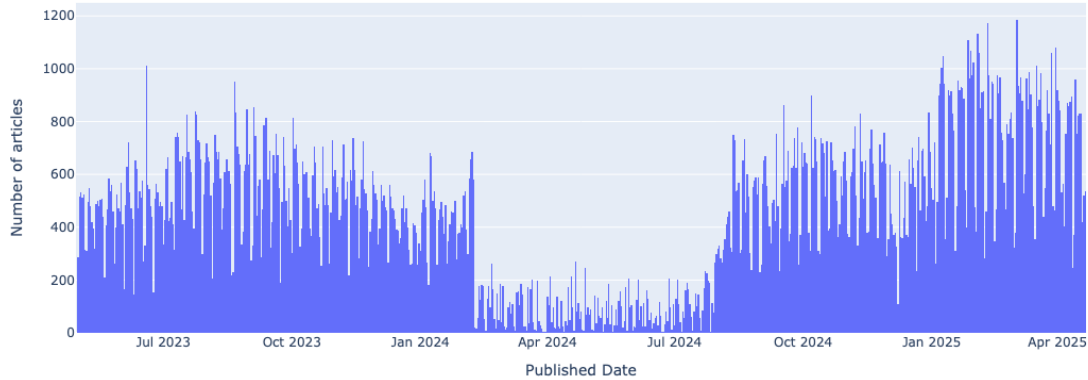


Figure 2.1: Number of labeled articles.

■ 2.4 Annotation Process

The quality labels used as the ground truth in this supervised learning task were assigned by experienced human editors at Seznam.cz. This annotation process is a routine part of content management on the platform, aimed at maintaining consistent editorial standards.

Editors evaluate articles based on a set of internal guidelines, but also rely significantly on their professional judgment and experience. They assess various aspects, including depth of content, writing quality, factual accuracy, structure, sourcing, and overall adherence to journalistic principles. Based on this assessment, each article is assigned one of the four quality labels defined in Section 1.2:

Class 0 (Insufficient Articles)

Class 1 (Short but Sufficient Articles)

Class 2 (Standard Articles)

Class 3 (High-Quality Articles)

■ 2.5 Dataset Statistics

■ Dataset Size and Splits

After cleaning and filtering, the data was split into three sets: Training Set, Validation Set and Test set. The distribution of articles across the four quality labels within each dataset split is presented in Tab. 2.1

Table 2.1: Class Distribution Across Dataset Splits

Set	Label 0	Label 1	Label 2	Label 3
Training	12,994 (22.9%)	32,086 (56.5%)	11,166 (19.7%)	542 (0.9%)
Validation	2,785	6,876	2,393	116
Test	2,795	6,863	2,404	118

As clearly visible from Tab. 2.1, the dataset exhibits significant class imbalance. The Short but Sufficient Articles constitutes the majority (over 56%) of the articles, while the

High-Quality Articles is extremely rare, representing less than 1% of the data. This imbalance presents a major challenge for model training, as standard algorithms tend to become biased towards the majority classes. Addressing this imbalance effectively is a key focus of the methodology described in the next chapter.

■ 2.6 Data Analysis

■ Article length

Figure Fig. 2.2 shows the marginal distribution of article length in words for each class. The average article length is 545 words.

The distribution is strongly right-skewed: the shortest admissible texts contain 51 words, the median is 434 words and the longest article reaches almost 9000 words. This long tail already hints at a qualitative difference between routine news briefs and in-depth reporting.

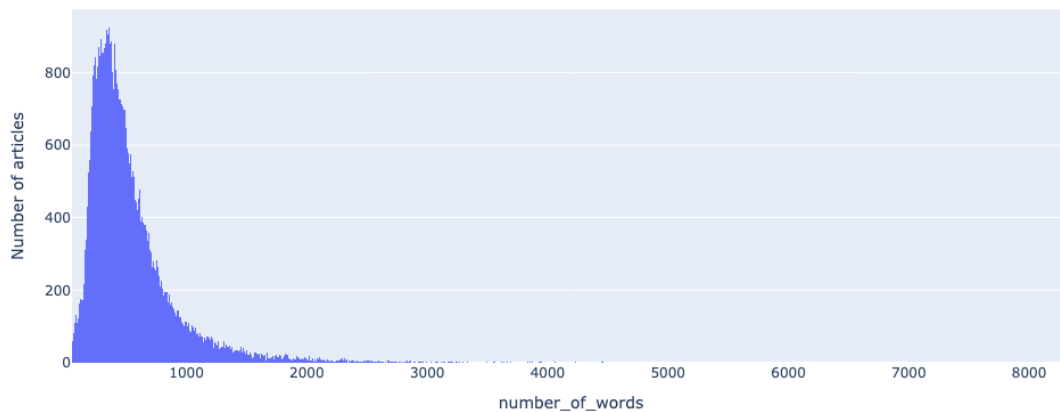


Figure 2.2: Distribution of article length in words.

■ Length by label

This Fig. 2.3 shows the distribution of article lengths for each quality label. The x-axis represents the number of words in an article, and the y-axis shows the probability density, so we can compare how article lengths differ across quality levels.

The results clearly show that better articles tend to be longer:

- Insufficient Articles and Short but Sufficient Articles articles are the shortest. Most of them are under 600 words, with label 1 forming the sharpest peak around 300-400 words.
- Standard Articles articles are longer, typically ranging from 400 to 1 200 words. They overlap with label 1 but clearly shift toward the right.
- High-Quality Articles articles are the longest. They have a wide distribution, with many texts over 1 500 words and some even exceeding 7 000 words. The long tail in the purple distribution reflects very detailed, in-depth articles.

This confirms that article length is a strong signal for quality. Longer texts are more likely to include detailed explanations, sources, and structured content—all features editors associate with high-quality journalism.

However, since the distributions overlap, especially between classes 1 and 2, article length alone is not enough to classify quality.

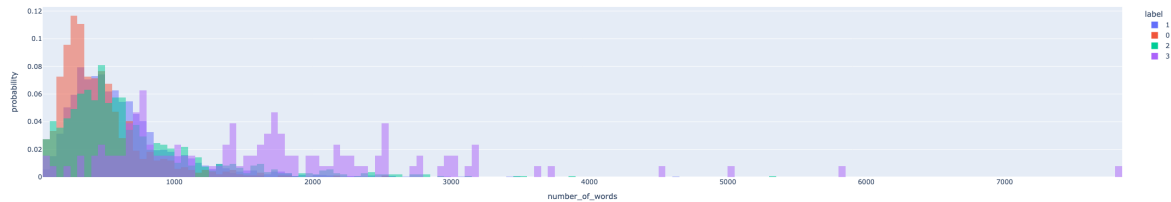


Figure 2.3: Distribution of article length in words by label

■ Unique-token probability density

The number of unique words in each article helps show how rich and varied the language is. Figure Fig. 2.4 shows that higher-quality articles usually have more unique words.

Insufficient Articles often have between 280 and 380 unique words and rarely go above 800. Most articles in Short but Sufficient Articles also stay under 500 unique words. Standard Articles articles use more words, and High-Quality Articles articles often have over 1000 unique words. Some even go above 3000.

This means that better articles tend to use a wider vocabulary. However, there is still a lot of overlap between the groups, especially for shorter texts. So, the number of unique words alone cannot perfectly tell the article quality but is still a useful signal. It works best when combined with other features like article length and sentence structure.

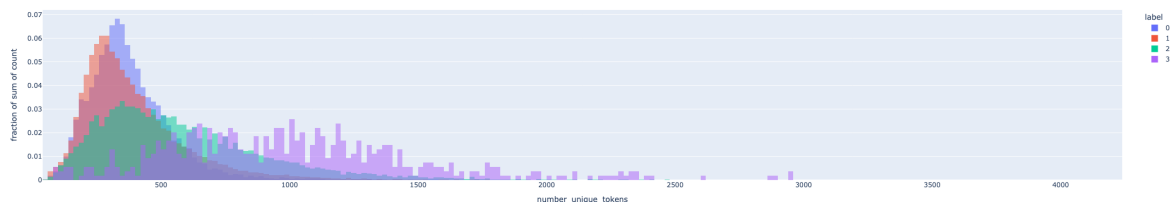


Figure 2.4: Distribution of unique tokens in articles.

■ 3 Evaluation Metrics

Evaluating the performance of a multi-class classification model for journalistic article quality requires careful selection of metrics, especially given the class imbalance in the dataset. As described in the Section 2.5, the four quality categories are not equally represented. This severe imbalance means that naive models could achieve seemingly high performance by favoring the majority class. Evaluation metrics must be chosen to reflect performance across all quality levels, rather than be dominated by the majority class.

■ 3.1 Limitations of Accuracy

Accuracy measures the proportion of correct predictions out of all predictions. While it is a simple and intuitive metric, accuracy is inadequate for imbalanced datasets. In our context of quality classification, accuracy treats all prediction errors uniformly and thus can be misleading. A model labeling every article as Short but Sufficient Articles would achieve 54% accuracy by virtue of Class 1's prevalence, potentially outperforming more nuanced models that try to distinguish all four classes but incur some mistakes on the majority class. This illustrates a key limitation: **accuracy can be artificially high even if the model entirely ignores minority classes.**

For a journalistic quality assessment system, such behavior is unacceptable – correctly identifying the minority classes (e.g. truly high-quality pieces) is crucial. Therefore, relying on accuracy alone would not highlight a model's failure to detect rare yet important cases. In light of this, we turn to alternative metrics that better capture performance across all classes, giving due weight to both common and uncommon article categories. These metrics focus on the balance between types of errors, rather than just the overall fraction correct.

■ 3.2 Precision

Precision is calculated as:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad (3.1)$$

where True Positive (TP) is the number of correctly predicted instances of a class, and False Positive (FP) is the number of incorrectly predicted instances where the model predicted the positive class but the actual label was negative.

Precision answers the question: “When the model predicts a certain class (especially a positive or high-quality outcome), how often is it correct?” In the context of our highest class (Class 3, high-quality articles), a high precision means that whenever the model flags an article as high-quality, it is usually correct. This is crucial for a platform like Seznam.cz because high precision in identifying top-quality content ensures that any article highlighted or promoted as “high-quality” truly meets the standards. In other words, high precision protects against false positives – it avoids promoting lower-quality articles as if they were premium content.

■ 3.3 Recall

Recall measures the fraction of actual positive instances correctly identified by the model, calculated as:

$$\text{Recall} = \frac{TP}{TP + FN}, \quad (3.2)$$

where False Negative (FN) is the number of actual positive instances that were incorrectly predicted as negative. It addresses “How many of the actual instances of a class did the model successfully find?” For High-Quality Articles, high recall means the model manages to identify most of the truly high-quality articles in the dataset. This is equally critical for our application: a high recall ensures that valuable content is not overlooked. In terms of business impact, high recall minimizes false negatives – it reduces the chance that an excellent article slips through the cracks and fails to be recognized or showcased as such. Missing a high-quality piece (a false negative) is a lost opportunity for user engagement and platform value, since exceptional content might not reach the audience it deserves.

■ 3.4 F1 Score

The balance between precision and recall is a classic trade-off: improving one can often degrade the other. In our case, both are important. To illustrate:

High Precision – Ensures that the articles labeled as high-quality are truly valuable, avoiding the risk of promoting subpar content to users.

High Recall – Ensures that most high-quality articles are detected and not ignored, maximizing the coverage of valuable content.

Depending on the application needs, one might prioritize precision over recall or vice-versa. For example, if the platform strictly wants to avoid ever featuring a low-quality article as top-tier content, it would emphasize precision (even if some good articles are missed). Conversely, if the goal is to surface as many great articles as possible, it would emphasize recall (even if a few lower-quality ones are misclassified among them). In this thesis, however, we aim for a balance, since both types of errors have costs (as discussed later).

To evaluate model performance holistically, we employ the **F1 score**, which is the harmonic mean of precision and recall. The F1 score provides a balanced measure that only yields a high value when both precision and recall are high, making it well-suited for our imbalanced multi-class problem. It is defined as:

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (3.3)$$

This formulation accounts for the trade-off between false positives (FP) and false negatives (FN): a model cannot maximize F1 by excelling in only one of precision or recall at the expense of the other.

Unlike accuracy—which, as noted, treats all errors the same and can be dominated by the majority class—the F1 score ensures that performance on minority classes is not masked. A high F1 indicates the model is catching most of the high-quality articles (high recall) and not mislabeling too many low-quality articles as high-quality (high precision). In practical terms, F1 was used as the primary metric during model development: for example, the training procedure monitored the F1 score on the validation set and employed early stopping when it ceased improving, ensuring the final model is the one that best balances precision and recall.

■ 3.5 Weighted F1 Score

When extending precision/recall/F1 to a multi-class setting, one must decide how to aggregate performance across classes. In this project we report the weighted F1 score as the primary evaluation metric, given the differing frequencies of each quality class. The weighted F1 is computed by calculating the F1 for each class separately, then taking a weighted average of these F1 scores using the class's support (proportion of samples) as weights. Formally:

$$\text{F1W} = \sum_{i=1}^n (w_i \times \text{F1}_i) \quad (3.4)$$

where $i \in \{0, 1, 2, 3\}$, $w_i = \frac{N_i}{N}$ is the fraction of samples belonging to class i in the dataset (i.e. its empirical probability) and F1_i is the F1 score for class i . This approach gives more influence to classes with more samples, while still incorporating performance on all classes.

■ Why use weighted F1?

In an imbalanced multi-class scenario like ours, weighted F1 offers several advantages over other averaging schemes (such as macro or micro averaging):

- **Balances Class Influence:** Macro-averaged F1 treats all classes equally, which can over-emphasize performance on very rare classes, while micro-averaging is dominated by the majority class (since it effectively counts every instance equally, making the aggregate metric similar to accuracy in a single-label setting). The weighted F1 is a compromise that considers each class's importance proportional to its frequency, preventing both the over-domination of majority classes and the over-sensitivity to tiny classes.
- **Comprehensive Single Metric:** The weighted F1 provides a single summary statistic that still reflects precision-recall balance across all categories. This is convenient for model selection and comparison. During development, we used the weighted F1 on the validation set to pick the best model and tune hyperparameters (see Methodology chapter on training procedure), since it ensured we were improving performance in an overall sense, not just for one favored class.
- **Reflects Real Data:** Weighted F1 aligns evaluation with the actual class distribution, ensuring that minority classes are considered without letting rare classes overly affect the result.

■ 3.6 Confusion Matrix

The confusion matrix is a powerful tool for visualizing the performance of a multi-class classification model. It provides a detailed breakdown of how many instances from each true class were predicted as each possible class. This allows us to see not only the overall accuracy but also where the model is making specific errors. Example of Confusion Matrix in Fig. 3.1.

In summary, the weighted F1 score was deemed the most appropriate metric for this study, and it is the primary metric reported for our results. It directly influenced model training as well: for instance, the final model was selected based on the highest validation weighted F1, and training was stopped early once this metric stopped improving to avoid overfitting. By using weighted F1 consistently, we maintain evaluation consistency between training and testing phases in the face of class imbalance.

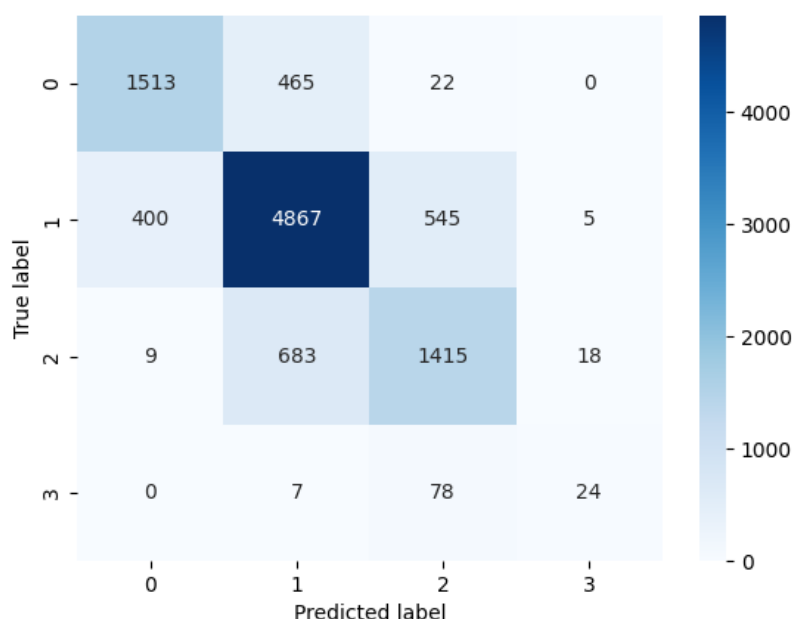


Figure 3.1: Example of Confusion Matrix. DebertaV3 Large.

■ 3.7 Business Relevance

The choice of evaluation metrics is not only a technical consideration but also a business decision in the context of a platform like Seznam.cz. The ultimate goal of automating journalistic quality assessment is to have the model’s predictions align with human editors’ judgments across all quality categories, from Insufficient Articles up to High-Quality Articles. As described in the Methodology chapter, human editors currently provide the quality labels, and their agreement rate sets an upper bound on model performance. The metrics we choose should thus reflect how well the model replicates those editorial decisions.

In this regard, the weighted F1 score aligns well with business objectives. It measures the model’s accuracy in mirroring human judgment while accounting for the prevalence of each quality category. In effect, a high weighted F1 indicates that the model is making the right distinctions between article qualities in roughly the same proportions as the real content mix on the platform. This is important for maintaining consistent quality standards: the platform seeks to reliably identify top-quality content and flag insufficient content, just as an editor would, but with far less manual effort. By using weighted F1 (which, as explained, ensures reasonable performance on each class), we directly support the goal that no class of article quality is neglected by the automated system. In business terms, this means the automation will uphold the overall editorial policy — it won’t simply label everything as “average” to get a good accuracy, but will correctly spot the outstanding pieces and the sub-par ones according to their true frequency. A metric aligned with these needs helps guide model development towards an outcome that is valuable for the platform’s content curation aims.

■ Cost of Classification Errors

In the context of content management, different types of classification errors carry distinct consequences, each affecting the platform in unique ways:

- **False Positives (FPs)** When lower-quality articles are incorrectly labeled as high-quality, they may be promoted as premium content. This can undermine user trust, as readers may feel misled by the platform's recommendations. Over time, such errors could harm the platform's reputation, making it less reliable in the eyes of its audience.
- **False Negatives (FNs)** Conversely, when truly high-quality articles are not recognized as such, the platform misses opportunities to showcase valuable material. This can reduce user engagement, as exceptional content fails to reach the audience it deserves, potentially limiting the platform's ability to retain and attract readers.

The F1 score addresses these challenges by striking a balance between precision (avoiding false positives) and recall (avoiding false negatives). This balance is crucial for managing the trade-off between promoting quality content and maintaining credibility. Additionally, the weighted variant of the F1 score ensures that less frequent categories—such as high-quality or insufficient articles—are not overlooked, despite their smaller presence in the dataset. By giving appropriate consideration to these minority classes, the metric supports a more comprehensive evaluation of the model's effectiveness, aligning with the platform's broader quality objectives.

■ 4 Methodology

■ 4.1 Editor Analysis and Upper Bound

■ Why Human Editors Are Important

When it comes to deciding what makes a good article, human judgment is key. Editors look at many factors, such as the depth of the content, the quality of the writing, the language used, and whether the article follows journalism standards. At Seznam.cz, experienced editors review articles to ensure they meet high standards. They follow clear guidelines but also rely on their own expertise. This human approach is valuable because it picks up on details and context that machines might miss. However, it's not perfect—it can be hard to stay consistent, it takes time, and it requires a lot of effort. Understanding how humans evaluate articles, including both their strengths and weaknesses, helps us design automatic systems that can do a similar job. Editors don't just focus on simple things like grammar or structure. They also check bigger issues, like whether the facts are correct, the story makes sense, and the journalism is fair and honest. But quality isn't always clear-cut—some aspects depend on personal opinions, so editors don't always agree. This is especially true for articles that aren't obviously great or terrible. These differences aren't a flaw; they just show how complex it is to judge quality.

■ How We Studied Editor Agreement

To understand how much editors agree and to set a target for automatic systems, we carried out a detailed study. We selected 100 articles from our dataset, choosing a mix of topics, dates, and quality levels to get a fair sample. Three experienced editors reviewed each article independently and sorted them into four categories. They followed the same rules used every day at Seznam.cz.

■ What We Learned About Editor Agreement

The results of our study were surprisingly strong—better than we expected. All three editors gave the same quality rating to 70 out of the 100 articles, which means a 70% full agreement rate. Even more impressive, at least two out of three editors agreed on the rating for every single article, giving us a 100% partial agreement rate. Interestingly, there was never a case where all three editors completely disagreed by picking totally different ratings. This tells us that while editors don't always see things exactly the same way, their disagreements are small—usually just between nearby categories, not about the article's overall quality. This shows a solid level of consistency among human experts.

■ Setting the Upper Bound

The 70% full agreement rate gives us a clear target, or “upper bound,” for our automatic classification model. This is the level where even human experts fully agree, so it's a realistic goal for the system to aim for. It's also important to note that disagreements between editors were never drastic—they typically differed by just one category. This means that a reliable model should also aim to stay within this range. In practice, it's acceptable if the model's

predictions differ from human ratings by at most one category, since even experienced editors make mistakes of only ± 1 level. Ensuring this small margin of error helps maintain the system's trustworthiness and mirrors real-world human judgment.

■ 4.2 Baseline Model Performance

Before developing complex models, we established baseline classifiers to provide lower-bound reference points. Baseline models are simple heuristics that do not incorporate any deep linguistic understanding, but they set minimum performance thresholds that any useful model should exceed. We implemented two naive strategies - a Random Classifier (Sec. 4.2.1) and a Most Frequent Class Classifier (Sec. 4.2.2) - and measured their effectiveness on the dataset. We report primarily the weighted F1 score as our performance metric.

■ Random Classifier

The Random baseline assigns a quality label to each article by sampling from the training set's class distribution. In other words, it ignores the article text and simply guesses a label at random, but with probabilities proportional to the observed frequencies of each class in the training data. This strategy produces an expected performance near chance level. Indeed, in our dataset – which is highly imbalanced (only 0.9% of articles are High-Quality, while the majority belong to the middle categories) – the random classifier achieved a weighted F1 of approximately 0.27. This very low F1 confirms that a classifier must do far more than random guessing to be useful. The random baseline underscores the challenge of the task: given the small fraction of high-quality articles and the dominance of the Short but Sufficient Articles, unguided guessing will mostly fail to identify the minority classes. This also highlights that any effective model needs to incorporate article-specific information (beyond class frequencies) to improve performance.

■ Most Frequent Class Classifier

The second baseline always predicts the single most common class from the training data for every article. In our case, the majority class was the Short but Sufficient Articles, which comprised a large portion of the dataset (Section 2.5 details the class distribution). By labeling every article as Short but Sufficient Articles, this classifier exploits the class imbalance to maximize overall accuracy, while entirely ignoring content. Perhaps surprisingly, this simplistic strategy achieved a weighted F1 of around 0.43, significantly higher than the random baseline. The reason is that by always predicting the majority class, it correctly “classifies” all truly Short but Sufficient Article (recall for Short but Sufficient Articles is 1.0), which carry a lot of weight in the weighted F1 calculation. This result demonstrates how a skewed dataset can let a trivial predictor attain a deceptively decent aggregate score. However, the limitations of the most-frequent-class model are clear: it fails to recognize any minority class (Precision and Recall are 0 for minority classes). It cannot differentiate the subtle cues that distinguish high-quality journalism from mediocre content.

■ Discussion

The results of these baseline models shed light on key aspects of our classification challenge. The Random Classifier's minimal performance underscores the necessity of incorporating article-specific information to achieve meaningful outcomes, rather than depending on random predictions. In contrast, the Most Frequent Class Classifier's score of 0.43 demonstrates the pronounced effect of class imbalance, as it achieves a respectable result solely by leveraging the majority class. These baselines set clear standards for evaluating future models. Any advanced classifier—whether employing intricate feature engineering or sophisti-

cated machine learning techniques—must exceed these initial benchmarks. Beyond improving overall metrics, we seek models that excel at correctly classifying less frequent yet significant quality categories. This establishes a critical goal for further development: to create systems that capture the nuanced elements of journalistic quality and more closely align with human editorial assessments.

■ 4.3 Performance of Pre-trained Language Models

Transformer-based language models have revolutionized natural language processing across multiple languages, including Czech. This chapter examines the theoretical foundations, architectures, and practical applications of pre-trained language models for Czech text classification tasks, with special attention to models specifically designed for the Czech language and their performance characteristics.

■ Theoretical Foundations

Pre-trained language models represent a paradigm shift in natural language processing, moving beyond traditional lexical approaches to capture deeper semantic relationships within text. These models leverage transformer architectures introduced by **vaswani2017attention** to create contextual representations that have proven remarkably effective for downstream tasks like classification, retrieval, and generation.

■ Evaluation of Multiple Pre-trained Models

For this research, four pre-trained Bidirectional Encoder Representations from Transformers (BERT)-like (**devlin2019bert**) language models developed internally at Seznam were evaluated. The models – Base-sized Roberta using Wordpiece tokenization for Czech (BROWCzech), Retrieval-Oriented Masked Autoencoder (RetroMAE), Decoding-enhanced BERT with Disentangled Attention (DeBERTaV3) Base, and DeBERTaV3 Large – were fine-tuned to predict quality ratings for articles.

BROWCzech

Architecture BROWCzech is a monolingual Czech language model developed by Seznam.cz. It is based on the RoBERTa (**liu2019roberta**) architecture and was pre-trained on an extensive internal corpus of Czech webpages, encompassing diverse topics and writing styles. This extensive pre-training enables the model to capture the nuances of the Czech language effectively.

- Parameters: 109.5 million (23.4M in the embedding layer + 86.0M in the transformer layers)
- Layers: 12 transformer layers
- Hidden Dimensions: 768
- Input Tokens: Maximum of 512
- Tokenizer: Monolingual, supporting only Czech

Pre-Training The model was trained using the Masked Language Modeling (MLM) objective, where a portion of the input tokens is masked, and the model learns to predict these masked tokens. This training approach allows the model to develop a deep understanding of the language structure and semantics. The pre-training corpus consisted of a vast collection of Czech web pages, ensuring a comprehensive representation of the language used in various contexts.

Special Capabilities Due to its monolingual focus and extensive pre-training on Czech data, BROWCzech excels in tasks requiring a deep understanding of the Czech language. It

is particularly effective in scenarios where the text is exclusively in Czech, providing high-quality contextual embeddings that enhance performance in downstream tasks such as text classification and sentiment analysis.

RetroMAE

Architecture RetroMAE (xiao2022retromae) is a transformer encoder that integrates a specialized pre-training architecture for better document-level representations. It introduces an asymmetric encoder-decoder setup during pre-training. The encoder is a full-scale BERT-like transformer which produces a condensed vector representation of the input text (for instance, the [CLS] token embedding). The decoder is a lightweight one-layer transformer that uses this representation to reconstruct the original text. This design does not change the model’s structure at fine-tuning time – we still fine-tune the 12-layer encoder on the classification task – but the pre-training setup encourages the encoder to act as a powerful sentence/document encoder. RetroMAE uses a Czech tokenizer, so in architecture it is comparable to BROWCzech aside from the pre-training apparatus.

- Parameters: 130.0 million (43.9M in the embedding layer + 86.0M in the transformer layers)
- Layers: 12 transformer layers
- Hidden Dimensions: 768
- Input Tokens: Maximum of 512
- Positional Embedding: absolute
- Tokenizer: Monolingual, supporting only Czech

Pre-Training RetroMAE was pre-trained on a massive Czech text corpus (253 GB of news and web articles) with a novel masked autoencoding strategy geared towards retrieval. The key idea is to train the encoder to produce embeddings that capture the meaning of the entire text, which can be used to regenerate the text. To achieve this, RetroMAE applies different masking ratios to the encoder and decoder inputs: for each input, the encoder sees a moderately masked version (20% tokens removed) while the decoder sees a heavily masked version (50% tokens masked out). The encoder must output a representation that, when fed into the decoder along with the decoder’s masked input, allows the decoder to reconstruct the original sentence (recover the masked tokens). Additionally, RetroMAE employs an asymmetric model structure, with the encoder being a full transformer (to encode rich features) and the decoder being minimal (to force the burden of representation onto the encoder). This pre-training approach effectively teaches the encoder to compress each document’s essential content into its embedding. Beyond autoencoding, RetroMAE’s training in this project also included retrieval-based contrastive learning: the model was periodically tasked to bring embeddings of similar documents closer and push dissimilar ones apart. This two-pronged pre-training (reconstruction + contrastive objectives) leads to embeddings that are not only information-rich but also arranged in a semantic space useful for comparing texts. Indeed, the creators of RetroMAE report that it dramatically improves performance on retrieval benchmarks like BEIR and MS MARC, confirming that the encoder learns powerful semantic representations.

Special Capabilities Thanks to its training, RetroMAE excels at capturing document-level semantics and topical depth. It effectively encodes the “aboutness” of an article – the model was literally trained to compress an article’s meaning into one vector. For journalistic quality, this means RetroMAE can distinguish content-rich, in-depth articles from superficial

ones. For example, an article with thorough coverage of a subject (lots of pertinent details, background context, and related information) will produce a dense embedding that RetroMAE recognizes as distinct from that of a short news brief. This strength aligns with the “depth” dimension of quality: RetroMAE should be adept at flagging articles that lack substance (likely classifying them as lower quality) versus those with comprehensive coverage (higher quality). Its retrieval objective may also enhance detection of factual consistency – articles that cover similar events should have similar embeddings, so if an article’s content is anomalous or off-topic (potentially indicating lower factual accuracy or relevance), RetroMAE might more readily identify it. On the other hand, RetroMAE does not introduce special handling of language beyond Czech. It might not handle foreign names or code-switching as well as a bilingual model, and it relies on the same 512-token input limit as BROWCzech. Also, because its focus is on what is said (semantic content) more than how it’s structured, RetroMAE might be less sensitive to certain stylistic or structural quality signals (e.g., subtle differences in tone or the presence of formatting elements like citations). In practice, however, document embeddings can encode some structural indicators if they correlate with content. Overall, RetroMAE’s semantic prowess is expected to help on quality aspects like factual depth and topic relevance, complementing other models that emphasize linguistic or structural nuances.

DeBERTaV3 Base

Architecture DeBERTaV3 ([he2021debertav3](#)) is a transformer encoder that builds on the BERT family with two major innovations: disentangled attention and improved embeddings. In DeBERTaV3, each token is characterized by two vectors – one for its content (token identity) and one for its position – instead of a single combined embedding. During self-attention, the model can attend to content and position information separately, a mechanism known as **Disentangled Attention**. This enables a form of relative position encoding: the model doesn’t just rely on fixed absolute positions, but learns relative positional biases so it can better handle long-range dependencies and varied word order. The Base model also employs an extended vocabulary (in our case, a bilingual Czech-English tokenizer) which increases the embedding layer parameters. The bilingual tokenizer and vocabulary allow it to natively process both Czech and English text. This is particularly useful for Czech news, which often contain English names, technical terms, or quotes. In summary, DeBERTaV3 Base’s architecture offers a more expressive attention mechanism and cross-lingual lexical coverage, setting it apart from BROWCzech and RetroMAE which use standard attention and monolingual vocabularies.

- Parameters: 129.4 million (43.9M in the embedding layer + 85.5M in the transformer layers)
- Layers: 12 transformer layers
- Hidden Dimensions: 768
- Input Tokens: Pre-trained with sentences with max 512 tokens, generalizes up to 2048 due to relative position embeddings
- Tokenizer: Bilingual, supporting both Czech and English

Pre-Training DeBERTaV3 Base was pre-trained with a state-of-the-art strategy that combines ideas from Efficiently Learning an Encoder that Classifies Token Replacements Accurately (ELECTRA) and DeBERTa. Instead of the traditional MLM objective, it uses Replaced Token Detection (RTD) ([clark2020electra](#)) – the model is trained to distinguish real tokens from plausible but fake tokens generated by a small generator network. This task is more sample-efficient than MLM, yielding better learning of token distinctions. Additionally,

DeBERTaV3 introduced Gradient-Disentangled Embedding Sharing, a technique to allow the generator and discriminator (in the ELECTRA setup) to share token embeddings without conflicting gradients (avoiding a “tug-of-war” in embedding updates). By applying RTD with disentangled embeddings, DeBERTaV3 achieves faster and more effective pre-training than MLM-based model. The model was trained on a mixture of English and Czech text (a bilingual corpus), reflecting its dual-language design. This means it saw Czech news articles and possibly English texts during pre-training, enabling it to handle code-switching and borrowed foreign phrases. The disentangled attention from the original DeBERTa ([he2020deberta](#)) is retained, meaning the model also learns relative position representations during pre-training. Overall, this advanced pre-training regimen endows DeBERTaV3 Base with both multilingual understanding and robust language modeling, which have been shown to boost performance on a wide range of NLP tasks and even cross-lingual benchmark.

Special Capabilities DeBERTaV3 Base brings together several capabilities that are particularly relevant for assessing journalistic quality. First, its bilingual nature ensures that the presence of English names or excerpts in a Czech article will not confuse the model – it recognizes those tokens and their meanings. This supports the transparency dimension, as articles often mention international organizations or use English technical terms; DeBERTa can comprehend these in context and contribute to a fair quality assessment. Second, the disentangled attention and relative positional encoding give DeBERTa an edge in understanding text structure and long-distance relationships. News articles have important structural elements (e.g., key facts at the beginning, quotes and references throughout). DeBERTaV3 can better model such patterns, for instance understanding that a statement followed by a dash and a name is likely a quote attribution – a subtle structural cue linked to transparency and credibility. It is also less likely to be thrown off by long sentences or paragraphs, since relative positioning helps it maintain coherence over longer spans. These abilities align with the structure and coherence aspect of quality: DeBERTaV3 might more accurately evaluate whether an article is well-organized and whether information is appropriately placed. Third, the model’s ELECTRA-style pre-training (RTD) tends to sharpen its ability to catch anomalies or errors in text (since it learned to spot “implausible” tokens). This could translate into better detection of factual errors or inconsistent details, aiding the factual accuracy dimension. In essence, DeBERTaV3 Base is expected to perform strongly across multiple quality criteria – it handles language diversity, captures long-range context, and has a finely-tuned sense for language correctness. Its main limitation relative to its Large counterpart is capacity: as a base model, it may sometimes miss very subtle signals that a larger model could pick up, and it shares the same 512-token input limit (mitigated by relative encoding, but still in effect). Nonetheless, among base-sized models, DeBERTaV3 Base is the most advanced in this lineup.

DeBERTaV3 Large

Architecture The Large variant of DeBERTaV3 significantly scales up the Base model, enhancing its capacity for handling nuanced and complex language tasks. It is specifically optimized for scenarios requiring deeper semantic understanding and improved performance in demanding classification and cross-encoder tasks.

- Parameters: 361.4 million (58.6M in the embedding layer + 302.8M in the transformer layers)
- Layers: 24
- Hidden Dimensions: 1024

- Input Tokens: pre-trained with 512 tokens, generalizes up to 2048 via relative position encoding
- Vocabulary Size: bilingual Czech-English

Pre-Training Unlike the Base variant, DeBERTaV3 Large is trained with an increased model capacity, allowing it to capture subtler language nuances and richer contextual relationships. While maintaining the ELECTRA-style RTD objective, its larger model size provides more parameters to effectively differentiate authentic and artificially generated tokens, resulting in improved learning efficiency and deeper semantic representations. The training corpus remains bilingual, similar to the Base variant, but the enhanced parameter count enables superior assimilation of cross-lingual contextual clues and more robust modeling of long-form and nuanced texts.

Special Capabilities Compared to DeBERTaV3 Base, the Large model demonstrates markedly improved performance in tasks requiring detailed semantic comprehension and fine-grained textual distinctions. Its increased number of transformer layers and expanded hidden dimensions enhance its ability to resolve subtle language patterns, ambiguous contexts, and complex factual consistency scenarios. The larger model is better suited for detailed anomaly detection tasks, such as identifying intricate factual errors or subtle inconsistencies. Additionally, its superior long-range modeling capability due to enhanced relative positional encoding makes it particularly effective in evaluating coherence and structural quality of lengthy articles or documents. This positions DeBERTaV3 Large as the preferred choice for high-stakes journalistic quality assessments and intricate text classification tasks, surpassing the Base variant in overall accuracy and contextual sensitivity.

■ Model Selection

To compare the performance of the evaluated models, we trained them using the same loss function (described in detail below), identical features, and the same context window size. Hyperparameters for each model were carefully tuned to achieve the best possible validation performance. Below, we present the weighted F1 scores and the F1 scores for each quality class (F1₁–F1₄) on the test set.

Model	F1 ₁	F1 ₂	F1 ₃	F1 ₄	F1 Weighted
BROWCzech	0.727	0.752	0.656	0.359	0.711
RetroMAE	0.725	0.749	0.656	0.299	0.721
DeBERTaV3 Base	0.727	0.752	0.656	0.359	0.724
DeBERTaV3 Large	-	-	-	-	0.740

Table 4.1: Validation F1 scores for each model. F1₁–F1₄ correspond to classification quality for each label class.

DeBERTaV3 Large achieved the best weighted F1 score on the validation set, indicating that it is better suited for the quality classification task compared to the other models. Despite all models being trained under the same conditions, the increased model capacity and architectural improvements of DeBERTaV3 Large allowed it to generalize more effectively on unseen data.

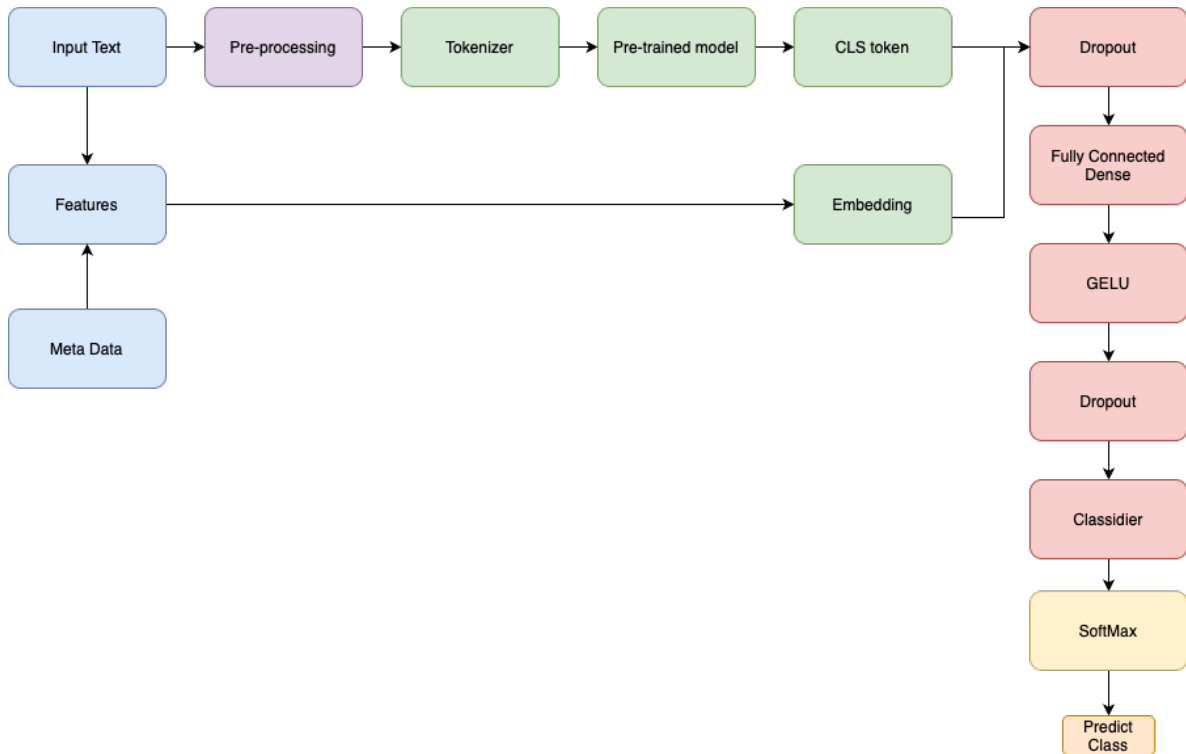


Figure 4.1: Model architecture overview for journalistic quality classification.

■ 4.4 Classifier Architecture Design

The classification architecture implemented in this research was designed to efficiently leverage the semantic information extracted by large pre-trained language models while introducing minimal additional complexity to prevent overfitting, particularly given the class imbalance of the dataset.

Initially, during experiments with the BROWCzech model, a simple classification head was developed, consisting of a dropout layer, a dense projection layer, and a softmax output. However, through systematic experimentation, it became evident that a slightly deeper and regularized architecture led to improved performance without significant computational overhead. These improvements, although first tested with BROWCzech, were later adapted and finalized in conjunction with the DeBERTaV3 model, which ultimately demonstrated superior performance and became the backbone of the final system.

The overall data flow and model architecture are illustrated in Figure 4.1. As shown, the input text is first pre-processed and tokenized, then passed through the pre-trained transformer model to obtain contextual embeddings. The [CLS] token embedding is processed through a regularized fully connected classifier head, including dropout, dense projection, GELU activation, and another dropout, before producing the final logits for prediction. Separately, additional features and metadata were extracted for possible use in ensemble models (described later in Section 4.9).

■ 4.5 Loss Function Engineering

Designing an effective loss function was critical due to the ordinal nature of the quality labels and the severe class imbalance in the dataset. The development of the loss function progressed through several stages, from a baseline categorical Cross-Entropy Loss (CE) to a final hybrid loss with an ordinal component and class weighting. Each refinement aimed to address specific shortcomings observed in earlier trials.

■ Initial Approach: Cross-Entropy Loss

The initial model training used the standard categorical CE loss, given its widespread success in multi-class classification. For a training sample i with true class label y_i and predicted probability $\hat{p}_{i,c}$ for each class c , the cross-entropy loss is defined as:

$$L_{\text{CE}} = -\frac{1}{N} \sum_{i=1}^N \log(\hat{p}_{i,y_i}), \quad (4.1)$$

where N is the number of training samples. This loss encourages the model to assign high probability to the correct class. CE was chosen as the starting point because it generally converges faster and more reliably than regression losses for classification problems (due to its strong gradients when the predicted probability is low for the true class). In early experiments, the CE loss yielded reasonable overall accuracy; however, closer inspection revealed two major issues: it **treated all misclassifications equally**, and it **struggled with the extreme class imbalance in the data**.

■ Shortcomings of Using CE Alone

Ordinal Errors Treated Equally: In our task, the four quality classes have an inherent order (from Insufficient Articles up to High-Quality Articles). A misclassification that is off by several levels (e.g. predicting class 0 for an article that is truly class 3) is far more severe than a misclassification to an adjacent class. Yet, standard cross-entropy provides no mechanism to differentiate these cases – it penalizes any incorrect prediction based only on the probability assigned to the true class, not how far off the predicted class is. In practice this meant the model might treat a confusion between class 3 vs 0 as no worse than between class 3 vs 2, despite the former being a much larger error in an ordinal sense. Such behavior is undesirable for quality assessment, where we want to strongly discourage large deviations in predicted quality. This limitation is well-recognized in ordinal classification literature: common classification losses “do not account for any distance information between prompting the introduction of specialized ordinal losses that penalize distant misclassifications more heavily.

Class Imbalance: Moreover, the dataset was highly imbalanced. Using plain cross-entropy, the model tended to optimize for the majority classes and could virtually ignore the minority class, since mis-predicting the rare class had minimal impact on the average loss. Indeed, an initial model with unweighted CE practically never predicted the High-Quality Articles, leading to an extremely low recall for that category (the model would default to predicting more frequent classes to minimize loss). This is a common outcome in imbalanced data settings, where the loss is dominated by the prevalent classes. Thus, additional measures were needed to ensure the model learned to recognize the minority class and to account for the ordinal scale of errors.

■ Addressing Class Imbalance: Focal Loss vs. Class Weighting

While the CE loss served as a strong initial baseline, we still needed to tackle the class imbalance so that minority classes (especially High-Quality Articles) would be properly recognized. Two strategies were explored: **Focal Loss** and **class weighting** within the cross-entropy.

Focal Loss Attempt

Focal loss was considered because it is specifically designed by **lin2017focal** to address class imbalance by down-weighting easy, majority-class examples. Focal loss adds a modulating factor $(1 - p)^\gamma$ to the standard CE loss, so that well-classified examples (with high p for the true class) incur much smaller loss, focusing training on the hard, misclassified examples. For a given sample i , the focal loss (without class-specific weighting) is:

$$L_{\text{FL}} = -\frac{1}{N} \sum_{i=1}^N (1 - \hat{p}_{i,y_i})^\gamma \log(\hat{p}_{i,y_i}), \quad (4.2)$$

where $\gamma \geq 0$ is the focusing parameter. In theory, this loss would make the model devote more attention to the High-Quality Articles, since those examples were consistently mispredicted (i.e. $\hat{p}_i, y_i$ for class 3 was often low, making $(1 - \hat{p}_i, y_i)^\gamma$ large and thus boosting their loss contribution). Focal loss has seen success in extreme imbalance settings (e.g. one-stage object detectors where background examples vastly outnumber foreground objects).

Outcomes with Focal Loss:

In our setting, however, focal loss did not yield the desired improvements. The model trained with focal loss showed lower overall F1 than the simpler weighted loss approach (described in Section 4.5.3), and it especially underperformed on the minority class. Several factors may explain this result.

First, the imbalance in our data, while significant, is not as astronomical as in some detection problems – thus the benefit of focal modulation was less pronounced. Second, focal loss can implicitly down-weight not just easy majority-class examples but also easy examples of all classes; this means some well-classified minority instances might receive less focus than they deserve. In practice, we found that the model still struggled to identify High-Quality articles with focal loss – presumably because those samples were extremely rare to begin with, and focal loss alone cannot increase their presence in the training signal enough. Tuning γ did not reverse this trend, and the added complexity of focal loss did not justify itself for our problem. Similar observations have been noted in literature when using focal loss outside its original context – if the class imbalance is better handled by explicit re-weighting, focal loss can sometimes even degrade performance by over-focusing on a handful of difficult cases and under-training on the rest. Given these findings, we turned to a more direct approach: class-weighted loss, which ultimately proved more effective.

Class Weighting:

The final loss function incorporated class weights into the cross-entropy term to directly counteract the imbalance. We assigned a weight w_c to each class c in the cross-entropy calculation, so that errors on minority classes are amplified and errors on majority classes are

down-weighted. The weights were set inversely proportional to class frequencies (with some manual tuning). In the end, the chosen weights were:

- $w_0 = 1.0$ for class 0: Insufficient Articles
- $w_1 = 0.5$ for class 1: Short but Sufficient Articles
- $w_2 = 1.0$ for class 2: Standard Articles
- $w_3 = 20.0$ for class 3: High-Quality Articles

These values mean, for example, that misclassifying a High-Quality Articles incurs $20\times$ the loss of misclassifying a Standard Articles, reflecting the high importance and rarity of Class 3. The weight for Class 1 was set to 0.5 because Short but Sufficient Articles was the most over-represented category (so its loss is halved to prevent the model from over-focusing on it). The weighted cross-entropy loss thus became

$$L_{CE}^* = -\frac{1}{N} \sum_i w_{y_i} \log \hat{p}_{i,y_i}. \quad (4.3)$$

■ Incorporating an Ordinal Component: CE + MSE

To address the **equal-penalty problem for ordinal errors**, the loss function was augmented with a Mean Squared Error (MSE) term on the class predictions. The idea was to introduce a penalty based on the distance between the predicted class and true class, treating the class labels as numerical values on an ordinal scale. In effect, this gives a harsher penalty to predictions that are far from the true label. Such a strategy aligns with approaches in other ordinal classification domains, where a class distance penalty is added to the loss to handle ordered labels. Formally, let the classes be encoded as ordinal values $\{0, 1, 2, 3\}$ for (Insufficient, Short, Standard, High-Quality). For a given sample i , define the predicted expected class (a single continuous value) as:

$$\hat{y}_i = \sum_{c=0}^3 c \cdot \hat{p}_{i,c}, \quad (4.4)$$

which is the expectation of the class index under the model's predicted distribution $\hat{p}_{i,c}$. Here $c = 0$ corresponds to Insufficient Articles and $c = 3$ to High-Quality Articles. The ground-truth label y_i is likewise taken as an integer in $\{0, 1, 2, 3\}$. The mean squared error for this sample is $(\hat{y}_i - y_i)^2$, and the MSE loss over the dataset is:

$$L_{MSE} = \frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i)^2. \quad (4.5)$$

This term grows quadratically with the difference between predicted and true class indices. For instance, if an article of true class High-Quality Articles is predicted as class Insufficient Articles, the term $(\hat{y} - y)^2$ will be much larger than if it were predicted as Standard Articles. By summing an MSE term with the cross-entropy, the hybrid loss function encourages correct classification and additionally encourages predictions to be closer to the true class when misclassifications occur. The combined loss was formulated as:

$$L_{\text{hybrid}} = L_{CE}^* + \lambda L_{MSE}, \quad (4.6)$$

where λ controls the relative importance of the ordinal MSE term. This approach effectively blends classification and regression objectives. The expectation was that the cross-entropy

component would drive the model to predict the correct class, while the MSE component would make it more sensitive to the magnitude of errors, thereby penalizing wildly incorrect predictions disproportionately. Literature on ordinal classification supports this intuition: a loss that “penalizes misclassifications more severely when they diverge farther from the true class” can improve performance on ordered-label tasks.

Balancing the Loss Terms (Choosing λ)

A key step was determining an appropriate value for λ . If λ is too low, the MSE term would be negligible and the loss reduces to standard CE (failing to use the ordinal information), if λ is too high, the model might overly optimize the MSE (treating the task like regression) at the expense of classification accuracy. To calibrate λ , the average magnitudes of the two loss components were analyzed. After an initial training epoch, the expected values of each loss were computed:

$$\mathbb{E}[L_{\text{CE}}^*] = \frac{1}{N} \sum_{i=1}^N L_{\text{CE}}(i), \quad \mathbb{E}[L_{\text{MSE}}] = \frac{1}{N} \sum_{i=1}^N L_{\text{MSE}}(i). \quad (4.7)$$

Their ratio

$$\alpha =; \frac{\mathbb{E}[L_{\text{CE}}]}{\mathbb{E}[L_{\text{MSE}}]} \quad (4.8)$$

provides a rough scaling factor to make the two loss terms contribute equally on average. In our case, this ratio α was used as an initial guess for λ . (Intuitively, if cross-entropy loss values were, say, ten times larger than MSE values, one would set $\lambda \approx 10$ to balance them.) Starting from $\lambda \approx \alpha$, we then experimented with a range of λ around this value to fine-tune performance.

Experiments with MSE Weight Several models were trained with different MSE weight values to evaluate their impact. Table 4.2 summarizes the weighted F1 results (the primary evaluation metric) for a range of λ values on the validation set:

MSE Weight λ	Weighted F1 (Validation)
0	0.66
0.5	0.68
1.0	0.702
2.0	0.693
3.0	0.69

Table 4.2: Performance for different MSE loss weightings.

As shown, the inclusion of the MSE term yielded a noticeable improvement over using pure cross-entropy ($\lambda = 0$). Performance peaked around $\lambda \approx 1.0$, which indeed was close to the calculated α in our case, confirming that balancing the two objectives was beneficial. A small MSE contribution (0.5) already improved the model’s F1, and using equal weight ($\lambda = 1.0$) for CE and MSE gave the highest F1. Increasing λ beyond 1.0 did not help further; a very large MSE weight started to slightly degrade performance (likely because the model then over-emphasized predicting the exact ordinal value, at times sacrificing classification fidelity). The best model with $\lambda = 1.0$ demonstrated that the ordinal loss component successfully reduced large misclassification errors – for example, the confusion matrix showed far fewer

cases of confounding the lowest vs. highest quality classes once the MSE term was introduced, compared to the CE-only loss. Our final loss function combined this weighted CE with the MSE term:

$$L_{\text{final}} = \frac{1}{N} \sum_{i=1}^N (w_{y_i} (-\log \hat{p}_{i,y_i}) + 1.0 \times (\tilde{y}_i - y_i)^2). \quad (4.9)$$

■ Final Results and Advantages of the Hybrid Loss

The $CE + 1.0 \times MSE$ with class weighting configuration was selected as the final loss function. This engineered loss function substantially outperformed the initial approaches. It effectively balanced the model’s performance across all classes, yielding a weighted F1 score of 0.702 on the test set – a significant improvement over the CE-only baseline. Notably, the recall for the High-Quality Articles class improved markedly; the model was able to identify a considerable portion of the High-Quality Articles that were previously being missed. This can be attributed to the $20\times$ class weight forcing the model to pay attention to class 3, in combination with the ordinal MSE term which dissuaded the model from conflating class 3 with much lower classes. In other words, the final loss configuration addressed both major challenges: it distinguished large vs. small misclassification errors (due to the MSE term) and mitigated class imbalance (due to the class weights on the CE term).

In summary, through iterative loss function engineering, the thesis arrived at a hybrid loss that leverages the strength of cross-entropy for classification, infuses ordinal awareness via a regression term, and incorporates strategic weighting to handle class frequency skew. This nuanced loss function was pivotal in boosting the model’s ability to mimic human-like judgments of article quality.

■ 4.6 Training Methodology

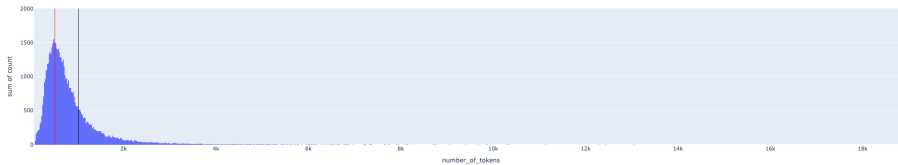


Figure 4.2: Token count per article

- 4.7 Training Stages
 - Hyperparameter Configuration
 - Validation and Early Stopping
- 4.8 Addressing Class Imbalance
 - Impact of Imbalance on Training
 - Class Weighting Strategy
- 4.9 Feature Engineering
 - 4.10 CatBoost
 - 4.11 Per-Class Performance Analysis
 - 4.12 Error Analysis

■ 5 Experiments and Results

■ 5.1 Experimental Setup

Describe evaluation metrics Explain baseline models Detail hardware/software environment

■ 5.2 Model Comparison

Present results of different model configurations Compare performance across quality categories Analyze errors and confusions between classes

■ 5.3 Feature Importance Analysis

Identify most influential features Analyze how different feature groups contribute Discuss implications for journalistic quality assessment

■ 6 Discussion, Conclusion and Future Work

■ 6.1 Discussion

Interpret the results in context of journalistic quality Compare your approach with existing methods Discuss practical implications for Seznam.cz

■ 6.2 Conclusion

Summarize key findings Revisit research questions and contributions Assess overall success of the approach

■ 6.3 Future Work

Suggest improvements to the model Propose additional features or approaches Discuss potential applications beyond the current scope

■ 7 Conclusion

Summarize the achieved results. Can be similar as an abstract or an introduction, however, it should be written in past tense.

■ 8 References

■ A Appendix A